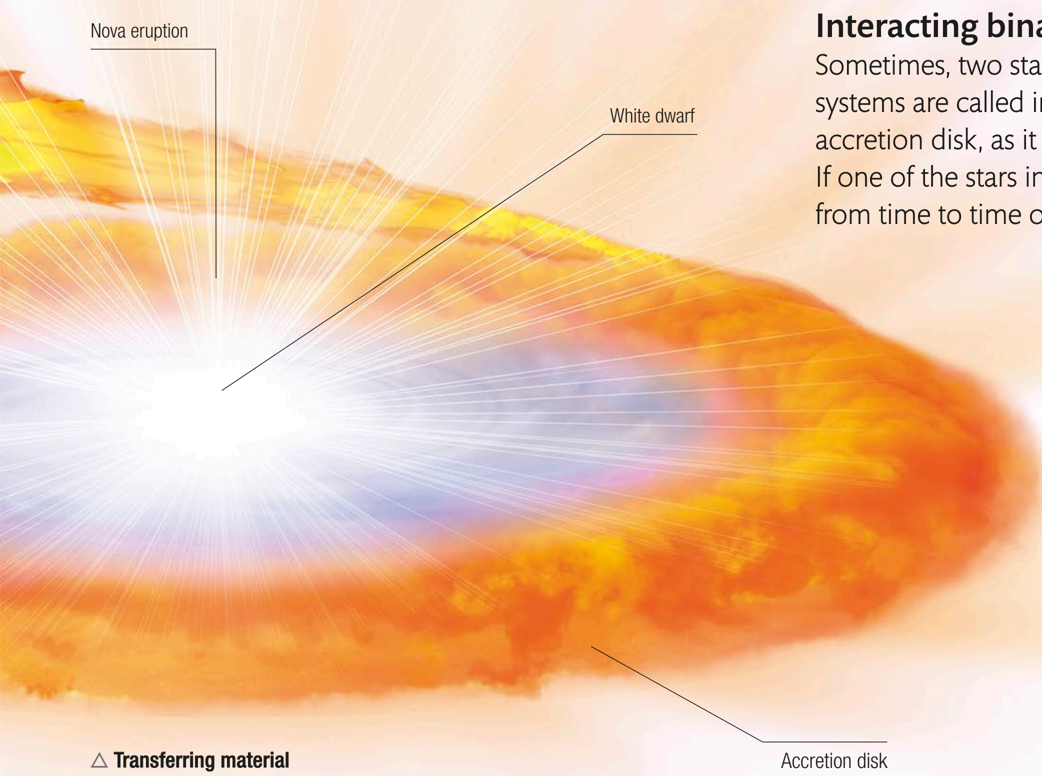
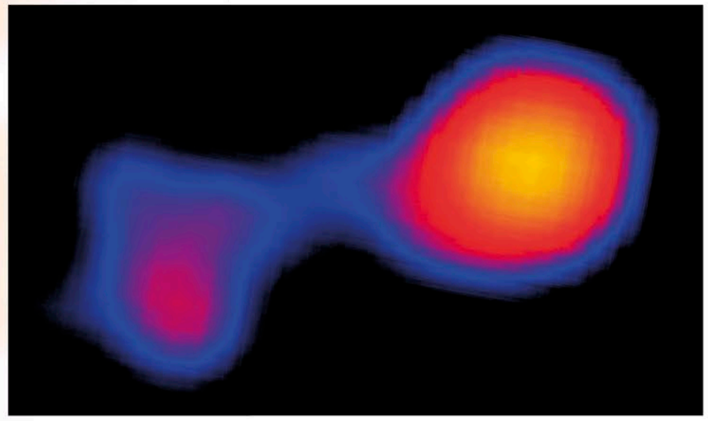
## Interacting binaries

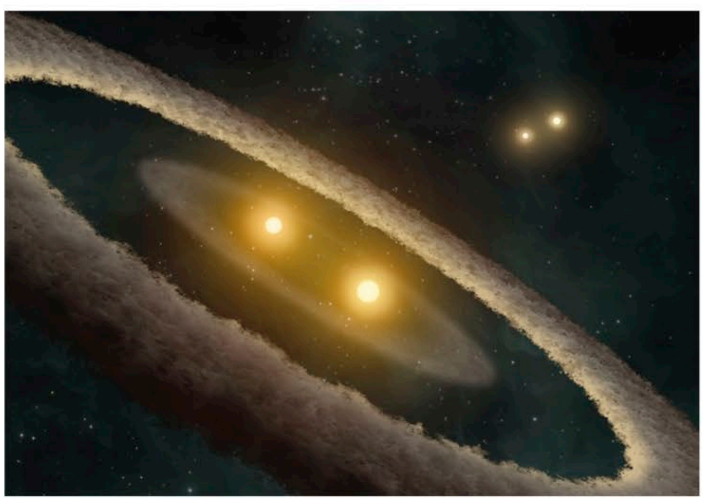
Sometimes, two stars are so close that material flows from one to the other. These systems are called interacting binaries. The transferred matter forms a disk, called an accretion disk, as it spirals in toward the receiving object. It may also release X-rays. If one of the stars in the binary is a white dwarf, explosions called novae may occur from time to time on the surface of the white dwarf.




△ **Transferring material**
In this interacting binary, a red giant orbits a white dwarf. Material from the giant is spilling onto the dwarf, forming an accretion disk with occasional nova outbursts.



△ **Mira system**
The star system Mira in the constellation Cetus consists of a red giant (which happens to vary in brightness) and a white dwarf, clearly separate in this X-ray image, with some material connecting the two stars.



▷ **HD 98800 system**
This artist's impression of the HD 98800 system shows two pairs of binary stars. All four stars are bound by gravity, but the distance between the two pairs is about 4.5 billion miles (7.5 billion km). A disk of gas and dust, with two distinct belts, surrounds one of the star pairs, and it is suspected that there is a planet orbiting in the gap between the belts.

◁ **True binary**
This telescope image clearly shows two bright stars—one gold, the other blue. The two stars are so close in the sky, however, that to the naked eye they look like a single star, which is known as Albireo (Beta Cygni). Astronomers think that Albireo's two components orbit each other, so they constitute a true binary, although each orbit takes about 100,000 years.

▷ **Di Cha system**
This complex star system, some 520 light-years away, contains four stars arranged in two pairs. Only the the two brightest are clearly visible in this Hubble Space Telescope image. However, all four stars are young and surrounded by a wispy wrapping of dust.